Compiler Construction

Lab 8

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BSCS-6

Task 1:

Lab 8 CC

Task: 1

$S \rightarrow AB$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid \varepsilon$

1) $first(A) = \{a, \varepsilon\}$ $\quad$ $follow(S) = \{\$\}$

$first(B) = \{b, \varepsilon\}$ $\quad$ $follow(A) = \{b, \$\}$

$first(S) = \{a, b, \varepsilon\}$ $\quad$ $follow(B) = \{\$\}$

2) ~~For~~ A:

$First(aA) = \{a\}$ $\quad$ $First(bB) = \{b\}$

$Follow(A) = \{b, \$\}$ $\quad$ $Follow(B) = \{\$\}$

Intersection $= \emptyset$ $\quad$ Intersection $= \emptyset$

$\{a\} \cap \{b, \$\} = \emptyset$ $\quad$ $\{b\} \cap \{\$\} = \emptyset$

Therefore grammer is LL(1).

Task 2:

Therefore grammer is LL(1)

Task: 2

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

$F \rightarrow id \mid (E)$

Input = id + id * id

Stack: $E

Input: id+ id*id$

Top = E (Non-terminal)

Replace = $E \rightarrow TE'$

Stack = $E'T

Date

Stack: $E'T

Top = T ← (Non-Terminal)

Replace = T → FT'

Stack = $E'T'F

Stack: $E'T'F

Top = F (Non-Terminal) (looking at grammer F→id | (E) input starts with id)

Replace = F → id

Stack = $E'T'id

Stack: $E'T'id

Top = id

Input = id

We remove both (from input & stack)

Stack: $E'T'

remaining input: +id*id $

Stack: $E'T'

Top = T'

Input = + (No *, so) use ε)

Stack = $E'

Stack: $E'

Input: +id*id $

Top = E'

Input = +

Use: E' → +TE'

Stack: $E'T+

Date

Stack : $ E' T +
Input = + id * id $
Top : + (Match)
Stack = $ E' T

Stack : $ E' T
Input = id * id $
Top = T
Replace = T → F T'
Stack : $ E' T' F

~~Stack~~ Stack : $ E' T' F
Input : id * id $
Top = F
Use = F → id
Stack : $ E' T' id

Stack : $ E' T' id
Input : id * id $
Top = id (Match)
Stack = $ E' T'

Stack : $ E' T'
Input = * id $
Top = T'
~~Input~~
Replace T → * F T'
Stack : $ E' T' F *

Stack : $ E' T' F *
Input = * id $
Top = * (Match)
Stack = $ E' T' F

Stack : $ E' T' F
Input = id $
Top = F
Replace F → id
Stack = $ E' T' id

Stack : $ E' T' id
Input : id $ ~~(Ft~~
~~Match id~~
Top : id (Match)
Stack : $ E' T'

Stack = $ E' T'
Top : T' , Input = $
Use T' → ε
Stack = $ E'

Stack = $ E'
Input = $
Top = E'
Use E' → ε

Stack = $ } String
Input = $ } accepted
nothing left in stack
nothing left in Input

Task 3:

task 3.cpp

```cpp
#include <iostream>
#include <stack>
#include <string>
using namespace std;


string input;
int i = 0;

void error()
{
    cout << "String Rejected (Syntax Error)" << endl;
    exit(0);
}

void match(char c)
{
    if(input[i] == c)
        i++;
    else
        error();
}

void S()
{
    if(input[i] == 'a')
    {
        match('a');
        S();
        match('b');
    }
    else if(input[i] == 'b' || input[i] == '\0')
    {
        return;
    }
    else
    {
        error();
    }
}

int main()
```

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```
Enter input: ab
String Accepted

--------------------------------
Process exited after 21.06 seconds with return value 0
Press any key to continue . . .
```

(globals)

task 3.cpp

```cpp
#include <iostream>
#include <stack>
#include <string>
using namespace std;


string input;
int i = 0;

void error()
{
    cout << "String Rejected (Syntax Error)" << endl;
    exit(0);
}

void match(char c)
{
    if(input[i] == c)
        i++;
    else
        error();
}

void S()
{
    if(input[i] == 'a')
    {
        match('a');
        S();
        match('b');
    }
    else if(input[i] == 'b' || input[i] == '\0')
    {
        return;
    }
    else
    {
        error();
    }
}

int main()
```

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```
Enter input: aabb
String Accepted

-----------------------
Process exited after 3.
Press any key to contin
```

task 3.cpp

```cpp
#include <iostream>
#include <stack>
#include <string>
using namespace std;


string input;
int i = 0;

void error()
{
    cout << "String Rejected (Syntax Error)" << endl;
    exit(0);
}

void match(char c)
{
    if(input[i] == c)
        i++;
    else
        error();
}

void S()
{
    if(input[i] == 'a')
    {
        match('a');
        S();
        match('b');
    }
    else if(input[i] == 'b' || input[i] == '\0')
    {
        return;
    }
    else
    {
        error();
    }
}

int main()
```

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```
Enter input: aaabbb
String Accepted

--------------------------------
Process exited after 4.995 seconds with return
Press any key to continue . . .
```

Step-by-step parsing process

Input string: ab

Derivation

S
-> aSb
-> aεb
-> ab

Step-by-step parsing

1. Start with S
2. Input starts with a, so apply **S -> aSb**
3. Match a
4. Now next symbol is b, so apply **S -> ε**
5. Match b

6. Input finished

Result

String Accepted